<u>Program -2</u>

Develop a program to Load a dataset with at least two numerical columns (e.g., Iris, Titanic). Plot a scatter plot of two variables and calculate their Pearson correlation coefficient. Write a program to compute the covariance and correlation matrix for a dataset. Visualize the correlation matrix using a heatmap to know which variables have strong positive/negative correlations.

Aim

To load a dataset with numerical attributes, plot a scatter plot between two variables, calculate the Pearson correlation coefficient, compute covariance and correlation matrices, and visualize the correlation matrix using a heatmap.

Theory

Correlation

Correlation measures the strength and direction of the relationship between two variables.

- **Positive Correlation** → both variables increase together.
- **Negative Correlation** → one variable increases while the other decreases.
- **Zero Correlation** → no relationship exists.

The Pearson correlation coefficient value lies between **-1 and +1**.

$r=\sum(xi-\bar{x})^2\sum(yi-\bar{y})^2/\sum(xi-\bar{x})^2(yi-\bar{y})^2$

Where:

- $r=1$ → perfect positive correlation
- $r=-1$ → perfect negative correlation
- $r=0$→ no correlation

Covariance

Covariance shows how two variables vary together.

- Positive covariance → variables move in the same direction.
- Negative covariance → variables move in opposite directions.
- $Cov(X,Y)=\sum(X_i-\bar{X})(Y_i-\bar{Y})/n-1$

Hatmap

A heatmap is a graphical representation of the correlation matrix where:

- Dark/high values indicate strong correlation.
- Light/low values indicate weak correlation.

Algorithm

1. Import required libraries.
2. Load the dataset.
3. Select two numerical columns.
4. Plot a scatter plot between the variables.
5. Compute Pearson correlation coefficient.
6. Compute covariance matrix.
7. Compute correlation matrix.
8. Visualize the correlation matrix using a heatmap.
9. Display the results.

[2]:
```python
import pandas as pd
# Load Titanic dataset
df = pd.read_csv('titanic_train.csv')

# Display first 5 rows
df.head()
```

[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

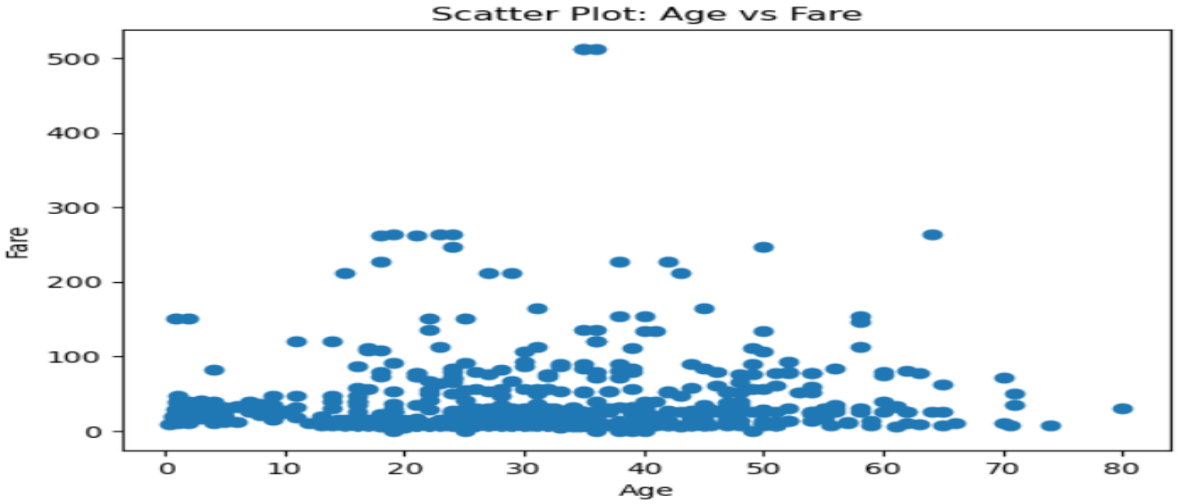
]:
```python
import pandas as pd
num_df = df[['Age', 'Fare', 'Pclass', 'SibSp', 'Parch']]
num_df = num_df.dropna()   # Remove missing values
num_df.head()
```

]:

	Age	Fare	Pclass	SibSp	Parch
0	22.0	7.2500	3	1	0
1	38.0	71.2833	1	1	0
2	26.0	7.9250	3	0	0
3	35.0	53.1000	1	1	0
4	35.0	8.0500	3	0	0

```python
import pandas as pd
import matplotlib.pyplot as plt

plt.figure()
plt.scatter(num_df['Age'], num_df['Fare'])
plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Scatter Plot: Age vs Fare")
plt.show()
```




```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('titanic_train.csv')
plt.figure(figsize=(8,6))
corr_matrix = num_df.corr()
sns.heatmap(corr_matrix, annot=True)
plt.title("Correlation Matrix Heatmap")
plt.show()
```

```python
import pandas as pd
df = pd.read_csv('titanic_train.csv')
cov_matrix = num_df.cov()
print("Covariance Matrix:")
print(cov_matrix)
```

```
Covariance Matrix:
               Age         Fare     Pclass     SibSp     Parch
Age     211.019125    73.849030  -4.496004 -4.163334 -2.344191
Fare     73.849030  2800.413100 -24.583138  6.806212  9.262176
Pclass   -4.496004   -24.583138   0.702663  0.052412  0.018370
SibSp    -4.163334     6.806212   0.052412  0.864497  0.304513
Parch    -2.344191     9.262176   0.018370  0.304513  0.728103
```

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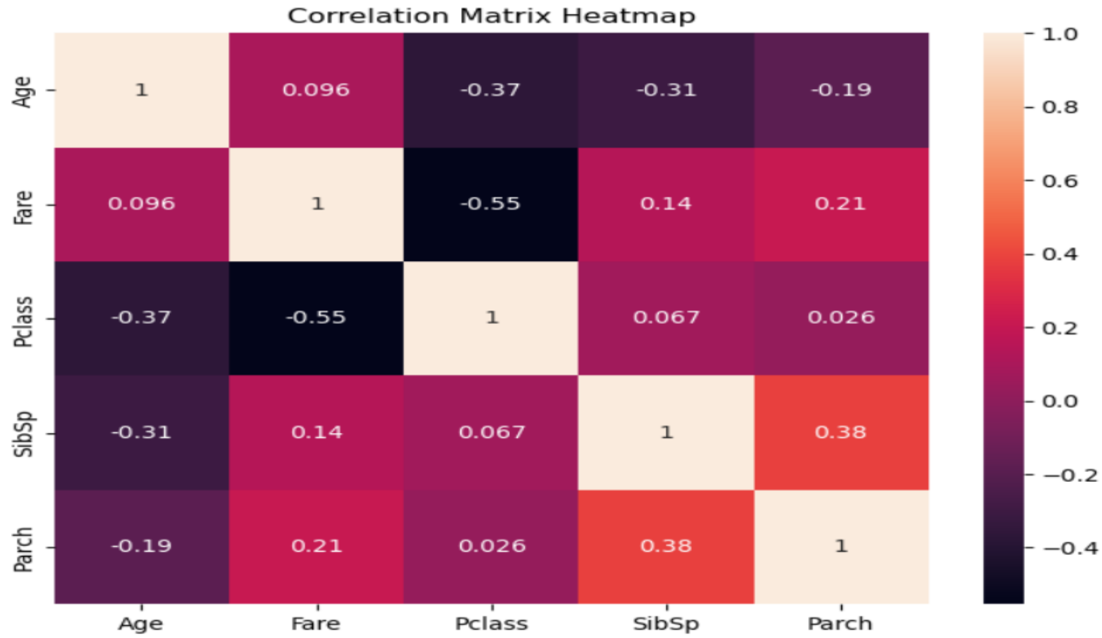

Result

Thus, the program to compute covariance and correlation matrices, calculate Pearson correlation coefficient, and visualize relationships using scatter plots and heatmaps was successfully implemented and executed.